(TR-104) PYTHON WITH AI/ML

Training Day 46 Report

Date: 16 March 2026

Topic Covered: RAG Pipeline Optimization, Retrieval Improvement & Production-Level Testing

The forty-sixth day of training focused on improving the reliability, retrieval quality, security, and production readiness of the RAG-based AI Assistant by optimizing chunking, retrieval filtering, guardrails, caching, and overall pipeline performance.

- Replaced PyPDFLoader with pdfplumber for improved PDF parsing and table extraction accuracy.
- Fixed incorrect company email information present in FAQ and HTML source files.
- Added similarity threshold filtering inside the retriever pipeline to remove irrelevant document chunks during retrieval.
- Increased minimum chunk size to improve semantic consistency of retrieved content.
- Improved FAQ chunk splitting logic using better separators for structured retrieval.
- Moved hardcoded LLM parameters and retrieval configurations into centralized settings and configuration files.
- Added vector database count functionality for easier monitoring and debugging of stored embeddings.
- Updated guardrail prompts to correctly allow company contact information, timings, and address-related responses.
- Fixed retrieval parameter inconsistencies inside the RAG pipeline configuration.
- Prepared a structured Alpaca-style dataset containing multiple company-related question-answer pairs for fine-tuning.
- Created a custom Ollama model configuration and integrated the techcorp-assistant model into the application workflow.
- Replaced the default base model configuration with the custom assistant model using environment variables.

- Updated the project knowledge base with a more structured company document for improved retrieval quality.
- Removed unused fine-tuning folders and cleaned the project structure for better maintainability.
- Updated requirements.txt with newly used dependencies for deployment consistency.
- Created a complete README.md file with detailed project setup instructions and documentation.
- Tested the RAG system on multiple company-related questions and verified correct response generation.
- Verified that out-of-scope questions correctly returned "I don't know" instead of hallucinated answers.
- Tested Redis caching and confirmed faster repeated query responses.
- Verified that guardrails successfully blocked prompt injection and malicious input attempts.
- Strengthened understanding of retrieval optimization, semantic chunking, caching, production-level guardrails, and deployment-ready RAG system architecture.
- GitHub Link: https://github.com/JaspinderKaurWalia26/rag_pipeline
- Blog Link: https://medium.com/@jaspinderkaurwalia855/building-a-production-ready-rag-pipeline-for-company-documents-using-local-llms-9d0c3300779e